

Lecture 11

SO(3), the double cover, and spherical harmonics

Objectives. Same algebra, second group. We prove that the adjoint map is a two-to-one surjection $SU(2) \rightarrow SO(3)$ with kernel $\{\pm\mathbb{I}\}$ — the double cover — and read off its topology: $SO(3)$ is the three-sphere with antipodal points identified, and it is not simply connected. We determine which spin representations descend (exactly the integer ones), completing Lecture 9's ledger; we realise them concretely as harmonic polynomials and spherical harmonics, proving $\dim = 2\ell + 1$, irreducibility, the Laplace eigenvalue $-\ell(\ell + 1)$ and completeness in $L^2(S^2)$; we give Euler angles and Wigner matrices the single page they deserve; and we resolve Lecture 10's IOU: quantum mechanics acts on rays, Wigner's theorem makes that precise, and the physicist's rotation group is therefore $SU(2)$.

11.1 Two groups, one algebra

Lecture 8 computed the Lie algebra of the rotation group: $\mathfrak{so}(3)$, the antisymmetric 3×3 matrices, with basis $(L_a)_{jk} = -\epsilon_{ajk}$ and brackets $[L_a, L_b] = \epsilon_{abc}L_c$ — the same structure constants as $\mathfrak{su}(2)$. The map

$$\psi: \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3), \quad \psi(X_a) = L_a,$$

is a Lie algebra isomorphism (Lecture 8), and Lecture 9 unmasked it: $\psi = \text{ad}$, the adjoint representation of $\mathfrak{su}(2)$ in the orthonormal basis $\{X_a\}$. Two consequences follow at once. First, the representation theory of the algebra $\mathfrak{so}(3)$ is already finished: transporting along ψ and complexifying (Lecture 10's lemma applies verbatim), the irreducible complex representations of $\mathfrak{so}(3)$ are the spin- j representations, one for every $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. Second, physics done infinitesimally cannot tell the two algebras apart: the Hermitian generators obey $[J_a, J_b] = i\epsilon_{abc}J_c$ on either side of ψ . Whatever distinguishes $SO(3)$ from $SU(2)$ is invisible to commutation relations.

What distinguishes them is global, and Lecture 9 taught us its name. The ledger of integration (Table 9.1) has a blank row: is $SO(3)$ simply connected, and which representations of its algebra integrate to the group? Today fills the row, and the instrument is the map ψ promoted from algebras to groups.

11.2 The double cover

Theorem 11.1 (The double cover). The adjoint map $\Phi := \text{Ad}: SU(2) \rightarrow SO(3)$, written in the orthonormal basis $\{X_a\}$ as in Lecture 9, is a continuous surjective homomorphism with

$$\ker \Phi = \{\pm\mathbb{I}\}.$$

Consequently Φ is exactly two-to-one — $\Phi(U) = \Phi(V)$ if and only if $V = \pm U$ — and Lecture 2's first isomorphism theorem gives

$$SO(3) \cong SU(2)/\{\pm I\}. \quad (11.1)$$

Proof. That Φ is a continuous homomorphism into $SO(3)$ is Lecture 9. Surjectivity: Lecture 9 also gives $\Phi(e^X) = e^{\text{ad}_X} = e^{\psi(X)}$ for $X \in \mathfrak{su}(2)$; the isomorphism ψ is onto $\mathfrak{so}(3)$, and Lecture 8 proved the exponential map onto $SO(3)$, so every rotation is $e^{\psi(X)} = \Phi(e^X)$. Kernel: $\Phi(U) = I$ means $UX_aU^{-1} = X_a$, i.e. U commutes with each Pauli matrix. Commuting with σ_3 forces U to preserve the eigenlines of σ_3 , hence to be diagonal, $U = \text{diag}(a, d)$; commuting with σ_1 , which swaps the two basis vectors, forces $a = d$; so $U = aI$ with $a^2 = \det U = 1$, giving $U = \pm I$. Finally $\Phi(U) = \Phi(V)$ iff $V^{-1}U \in \ker \Phi$ iff $U = \pm V$. ■

Remark 11.2 (A quotient that lands on its feet). Equation (11.1) is the only place Part II touches an abstract quotient group. Quotients of matrix groups need not be matrix groups — nothing in Lecture 2's construction produces matrices — and our framework demands closed matrix groups. Here we are lucky twice over: the quotient is not merely isomorphic to some group, it is concretely realised, by Φ itself, as the matrix group $SO(3)$. The abstract detour lasts one sentence.

Geometry first, then topology. Under the identification $SU(2) \cong S^3$ of Lecture 10, the fibres of Φ are the pairs $\{\pm U\}$ — antipodal points of the sphere. Since S^3 is compact and $SO(3)$ is Hausdorff, the continuous surjection Φ induces a continuous bijection from the quotient $S^3/\{\pm 1\}$ onto $SO(3)$, hence a homeomorphism. So as a metric space, $SO(3)$ is the three-sphere with antipodal points glued, each rotation owning exactly two antipodal addresses upstairs. The halved angles of Lectures 8–10 are this picture in coordinates (Figure 11.1): as θ runs from 0 to 2π , the rotation $R_z(\theta)$ walks a closed loop downstairs while its address $e^{-i\theta\sigma_3/2}$ walks only half of a great circle upstairs, from I to $-I$.

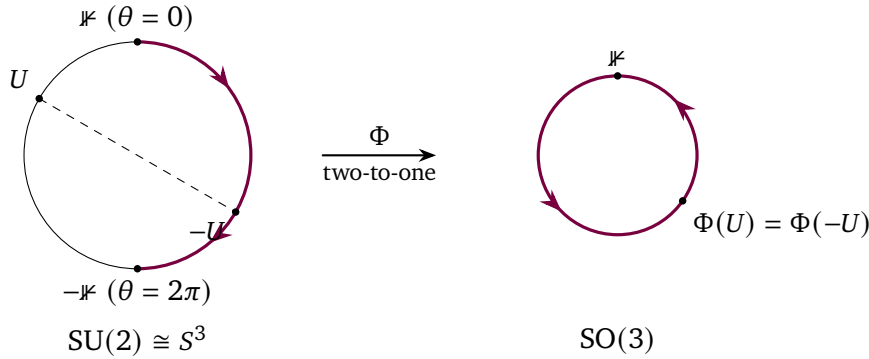


Figure 11.1: The double cover. Left: a great circle $\theta \mapsto e^{-i\theta\sigma_3/2}$ of $SU(2) \cong S^3$; as θ runs from 0 to 2π it traces the accented *open* path from I to $-I$ — half the circle, the halved angle at work. Right: its image under Φ , the *closed* loop $R_z(\theta)$ in $SO(3)$, traversed once in full. Antipodal points $\pm U$ (dashed chord) share a single image: Φ glues opposite points of the sphere. The open lift is the obstruction of Proposition 11.3 — contracting the loop downstairs would have to drag the lift's endpoint through the discrete fibre $\{\pm I\}$ — while doubling the range to 4π closes the lift, which then contracts in S^3 .

Proposition 11.3 (The loops of $SO(3)$). $SO(3)$ is connected (Lecture 8) but not simply connected: the loop $\rho(t) = R_z(2\pi t)$, $t \in [0, 1]$, is not contractible. Traversed twice, however, it is. (That every loop in $SO(3)$ shares this fate — the obstruction is exactly a sign — we state without proof.)

Proof (sketch; the patching follows Lecture 9). Doubling is the easy half. The loop ρ traversed twice is the Φ -image of $u(t) = e^{-2\pi i t \sigma_3}$, which is a genuine loop in $SU(2)$ ($u(0) = u(1) = \mathbb{1}$). Lecture 9 contracts u inside S^3 ; composing that homotopy with the continuous Φ contracts ρ^2 inside $SO(3)$, basepoint fixed.

For the other half, lift. Around any point of $SU(2)$ the map Φ is locally invertible — $\Phi(e^X) = e^{\psi(X)}$ together with Lecture 8's exponential charts shows it is a local homeomorphism with two continuous local inverses, differing by the sign of the sheet — so paths and homotopies in $SO(3)$ lift to $SU(2)$ by exactly the subdivide-and-patch argument that lifted circle loops in Lecture 9. The lift of ρ starting at $\mathbb{1}$ is $e^{-i\pi t \sigma_3}$, ending at $-\mathbb{1}$: the loop opens up upstairs. If a homotopy contracted ρ , its lift would move the endpoint of the lifted path continuously through the discrete fibre $\{\pm\mathbb{1}\}$ — hence not at all — while connecting it to the lift of the constant loop, whose endpoint is $+\mathbb{1}$. Contradiction. ■

Remark 11.4 (The belt trick). Proposition 11.3 can be held in the hands. Give a belt a full 2π twist while keeping both ends fixed: no amount of manoeuvring the strap (= homotopy of the path of frames) removes the twist. Give it 4π , and the twist can be undone — Dirac's belt trick (Figure 11.2). Notebook 11 animates both, next to the corresponding paths in $SU(2)$.

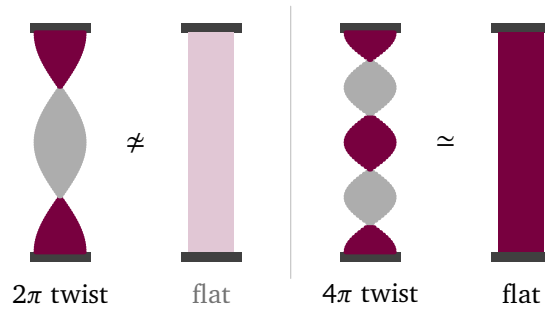


Figure 11.2: Dirac's belt trick, the topology of Proposition 11.3 held in the hands. A strap with both ends clamped records a path of frames by its twist; manoeuvring the strap without rotating the ends is a homotopy. A single 2π twist (left) cannot be removed — it is not homotopic to the untwisted strap, just as $\rho(t) = R_z(2\pi t)$ is not contractible in $SO(3)$. A 4π twist (right) *can* be worked out flat by passing the strap over the clamped end, exactly as ρ^2 contracts. The front of the strap is accented, its back grey, so each half-twist shows as a colour flip pinched to an edge.

So $SU(2)$ and $SO(3)$ share an algebra but not their loops, and by Lecture 9 that is precisely the difference that representation theory feels.

11.3 Which representations descend

Theorem 11.5 (Descent through the kernel). *Composition with Φ is a bijection between continuous finite-dimensional representations of $\text{SO}(3)$ and those representations Π of $\text{SU}(2)$ with $\Pi(-\mathbb{1}) = \mathbb{1}$, matching invariant subspaces and irreducibility. In particular, since $\Pi_j(-\mathbb{1}) = (-1)^{2j} \mathbb{1}$ (Lecture 10), the irreducible representations of $\text{SO}(3)$ are exactly the integer-spin ones,*

$$\Pi_\ell, \quad \ell = 0, 1, 2, \dots, \quad \dim \Pi_\ell = 2\ell + 1 = 1, 3, 5, \dots,$$

and no continuous representation of $\text{SO}(3)$ contains half-integer spin.

Proof. If $\tilde{\Pi}$ represents $\text{SO}(3)$, then $\Pi = \tilde{\Pi} \circ \Phi$ is a continuous representation of $\text{SU}(2)$ killing $\ker \Phi$; since Φ is onto, the two have the same invariant subspaces, and Π determines $\tilde{\Pi}$. Conversely, if $\Pi(-\mathbb{1}) = \mathbb{1}$ then Π is constant on the fibres $\{\pm U\}$, so there is a unique $\tilde{\Pi}$ with $\tilde{\Pi} \circ \Phi = \Pi$; it is a homomorphism because Φ is onto, and continuous because Φ has continuous local inverses (the continuous local inverses s of Φ , the *local sections* of Proposition 11.3's proof), so that locally $\tilde{\Pi} = \Pi \circ s$. Lecture 10's list and sign $(-1)^{2j}$ finish the classification. ■

Lecture 9's ledger closes: $\text{SO}(3)$ is *not* simply connected, and the algebra representations that integrate to it are exactly those built from integer spins. The algebra could not see this — ψ is an isomorphism — the loop legislated. And the verdict agrees with analysis: Lecture 9 derived integer quantisation of orbital angular momentum from single-valuedness on a circle of rotations; today's theorem is the same fact read globally, since function actions are honest $\text{SO}(3)$ representations and must therefore carry integer spin only. Where half-integers live we already know (Lecture 10); *why they are lawful* is Section 11.6. First, the integer representations deserve their concrete home.

11.4 Spherical harmonics: the integer representations, made of functions

Lecture 10 built every spin from polynomials in two complex variables. The integer spins admit a second construction, on the physicist's home ground: polynomials on $\mathbb{R}^3$, restricted to the sphere. Write (x, y, z) for coordinates on $\mathbb{R}^3$, $r = |x|$, and let

$$P_\ell = \{\text{homogeneous complex polynomials of degree } \ell \text{ in } x, y, z\}, \quad \dim P_\ell = \frac{(\ell+1)(\ell+2)}{2},$$

(count the monomials $x^a y^b z^c$ with $a + b + c = \ell$). The function action $(\Pi(R)f)(x) = f(R^{-1}x)$ of Lecture 3 preserves degree and acts continuously: a representation of $\text{SO}(3)$ on P_ℓ . One operator tells the spaces apart.

Lemma 11.6 (The Laplacian is a rotation invariant). *For every $R \in \text{O}(3)$ and smooth f , $\Delta(f \circ R^{-1}) = (\Delta f) \circ R^{-1}$.*

Proof. For a linear map A the chain rule gives $\partial_a(f \circ A) = \sum_b A_{ba} (\partial_b f) \circ A$, hence

$$\Delta(f \circ A) = \sum_{a,b,c} A_{ba} A_{ca} (\partial_b \partial_c f) \circ A = \sum_{b,c} (AA^T)_{bc} (\partial_b \partial_c f) \circ A,$$

and $AA^T = \mathbb{1}$ for orthogonal A . ■

Definition 11.7 (Harmonic polynomials). $H_\ell = \{f \in P_\ell : \Delta f = 0\}$. By Lemma 11.6, H_ℓ is an invariant subspace of P_ℓ .

How big is it? The answer drops out of an inner product tailored to differentiation — the same trick as Lecture 10's Gaussian, in differential clothing. For $f = \sum_{\alpha} c_{\alpha} x^{\alpha}$ (multi-index notation: $\alpha = (a, b, c)$, $x^{\alpha} = x^a y^b z^c$, $\partial^{\alpha} = \partial_x^a \partial_y^b \partial_z^c$, $\alpha! = a! b! c!$) write $\tilde{f}(\partial) = \sum_{\alpha} \tilde{c}_{\alpha} \partial^{\alpha}$, and on each P_{ℓ} define

$$\langle f, g \rangle_{\partial} = (\tilde{f}(\partial) g)(0). \quad (11.2)$$

Monomials are orthogonal with $\langle x^{\alpha}, x^{\alpha} \rangle_{\partial} = \alpha! > 0$, so (11.2) is an inner product; and since multiplying a polynomial by $r^2 = x^2 + y^2 + z^2$ corresponds, after the substitution $x \mapsto \partial$, to applying Δ , it obeys the adjunction

$$\langle r^2 f, g \rangle_{\partial} = \langle f, \Delta g \rangle_{\partial}, \quad f \in P_{\ell-2}, \quad g \in P_{\ell}.$$

Proposition 11.8 (Dimension of H_{ℓ}). $\Delta: P_{\ell} \rightarrow P_{\ell-2}$ is surjective, and

$$P_{\ell} = H_{\ell} \oplus r^2 P_{\ell-2} \quad (\text{orthogonal for (11.2)}), \quad \dim H_{\ell} = 2\ell + 1.$$

Iterating, $P_{\ell} = H_{\ell} \oplus r^2 H_{\ell-2} \oplus r^4 H_{\ell-4} \oplus \cdots$, so on the sphere ($r = 1$) every polynomial restricts to a sum of restricted harmonics.

Proof. If $h \in P_{\ell-2}$ is orthogonal to the range of Δ , the adjunction gives $\langle r^2 h, g \rangle_{\partial} = 0$ for all $g \in P_{\ell}$, so $r^2 h = 0$ and hence $h = 0$: Δ is surjective, and $\dim H_{\ell} = \dim P_{\ell} - \dim P_{\ell-2} = \frac{(\ell+1)(\ell+2) - \ell(\ell-1)}{2} = 2\ell + 1$. The adjunction also identifies $\ker \Delta$ with the orthogonal complement of $r^2 P_{\ell-2}$, giving the decomposition; iterate on the second summand. ■

The dimension $2\ell + 1$ is no coincidence, and the ladder machinery proves it. Assemble the generators for the three axes (Lecture 9 wrote $L_z = -i(x\partial_y - y\partial_x)$; L_x, L_y follow by cyclic permutation) into Hermitian operators and ladders,

$$L_z = -i(x\partial_y - y\partial_x), \quad L_{\pm} = L_x \pm iL_y = \pm z(\partial_x \pm i\partial_y) \mp (x \pm iy)\partial_z,$$

which satisfy Lecture 10's ladder brackets $[L_z, L_{\pm}] = \pm L_{\pm}$, $[L_+, L_-] = 2L_z$ — same structure constants, same theorems.

Theorem 11.9 (H_{ℓ} is the spin- ℓ representation). H_{ℓ} is irreducible, of spin ℓ , with highest-weight vector

$$w_{\ell} = (x + iy)^{\ell}.$$

Proof. First, w_{ℓ} lives in H_{ℓ} and is highest of weight ℓ :

$$\Delta(x + iy)^{\ell} = \ell(\ell - 1)(x + iy)^{\ell-2}(1^2 + i^2) = 0, \quad L_z w_{\ell} = \ell w_{\ell}, \quad L_+ w_{\ell} = 0,$$

the last two because $(\partial_x + i\partial_y)w_{\ell} = \ell(x + iy)^{\ell-1}(1 + i^2) = 0$ and $\partial_z w_{\ell} = 0$. Now run Lecture 10's descent inside H_{ℓ} from w_{ℓ} : the string formulas hold in any representation (their derivation in Lecture 10 used only the brackets and a highest-weight vector, not irreducibility), so $v_k = L_-^k w_{\ell}$ satisfies $L_z v_k = (\ell - k)v_k$ and $L_+ v_k = k(2\ell - k + 1)v_{k-1}$; the nonzero v_k are independent, and if $v_{N+1} = 0 \neq v_N$ then $0 = (N+1)(2\ell - N)v_N$ forces $N = 2\ell$ exactly. The span of $v_0, \dots, v_{2\ell}$ is therefore a $(2\ell + 1)$ -dimensional invariant subspace of H_{ℓ} , irreducible by the simple-spectrum ladder argument of Lecture 10's polynomial model — and by Proposition 11.8 it is H_{ℓ} . ■

Definition 11.10 (Spherical harmonics). $Y_{\ell m}$ is the restriction to S^2 of the basis vector of H_{ℓ} of weight m , normalised in $L^2(S^2)$ and phased by the ladder convention of Lecture 10, the one free overall phase per ℓ fixed by $Y_{\ell 0}(\hat{z}) > 0$ (the standard tables' choice). Concretely: $Y_{\ell \ell} \propto (x + iy)^{\ell}|_{S^2}$ and $Y_{\ell, m-1} \propto L_- Y_{\ell m}$.

For $\ell = 1$ the whole construction fits in one line:

$$w_1 = x + iy, \quad L_- w_1 = -2z, \quad L_-^2 w_1 = -2(x - iy), \quad L_-^3 w_1 = 0,$$

so $Y_{1,\pm 1} \propto (x \pm iy)|_{S^2}$ and $Y_{1,0} \propto z|_{S^2}$ — the p -orbitals of chemistry, generated rather than tabulated. Sheet 11 runs $\ell = 2$ and compares with the standard tables. Note also, free of charge from homogeneity: under the parity map $x \mapsto -x$ (which commutes with rotations), H_ℓ has the definite sign $(-1)^\ell$ — a fact Lecture 12's selection rules will spend.

So far the sphere has been a place to restrict polynomials. The restriction in fact carries all the structure: a homogeneous polynomial vanishing on S^2 vanishes identically (homogeneity: $u(x) = r^\ell u(x/r)$), so H_ℓ maps bijectively onto

$$\mathcal{Y}_\ell := \{u|_{S^2} : u \in H_\ell\} \subset L^2(S^2),$$

and the bijection intertwines the rotation actions. Two theorems make $\mathcal{Y}_\ell$ the centrepiece of mathematical physics: its elements diagonalise the sphere's Laplacian, and together they fill $L^2(S^2)$.

Definition 11.11 (The spherical Laplacian). For g smooth on S^2 , let $(Eg)(x) = g(x/|x|)$ be its degree-zero homogeneous extension, and define $\Delta_{S^2} g := (\Delta Eg)|_{S^2}$. (In spherical coordinates this is the familiar angular operator $\sin^{-1}\theta \partial_\theta (\sin \theta \partial_\theta) + \sin^{-2}\theta \partial_\varphi^2$; the verification is Notebook 11's.)

Theorem 11.12 (The eigenvalue $-\ell(\ell + 1)$). Every $Y \in \mathcal{Y}_\ell$ satisfies

$$\Delta_{S^2} Y = -\ell(\ell + 1) Y. \quad (11.3)$$

Proof. Write $Y = u|_{S^2}$ with $u \in H_\ell$; homogeneity gives $u = r^\ell EY$. For h homogeneous of degree zero, Euler's identity $(x \cdot \nabla)h = 0$; differentiate $h(tx) = h(x)$ at $t = 1$ kills the cross term in

$$\Delta(r^\ell h) = (\Delta r^\ell) h + 2 \nabla r^\ell \cdot \nabla h + r^\ell \Delta h = \ell(\ell + 1) r^{\ell-2} h + r^\ell \Delta h,$$

using $\partial_a r^\ell = \ell r^{\ell-2} x_a$, hence $\Delta r^\ell = \ell(\ell + 1) r^{\ell-2}$. Applying this to $h = EY$ and $u \in H_\ell$: $0 = \Delta u = \ell(\ell + 1) r^{\ell-2} EY + r^\ell \Delta(EY)$. Since $\Delta(EY)$ is homogeneous of degree -2 , restricting to $r = 1$ gives (11.3). ■

Remark 11.13 (Analysis meets algebra). The eigenvalue is Lecture 10's Casimir at $j = \ell$. Indeed $\mathcal{Y}_\ell$ is the spin- ℓ representation, on which $L^2 = L_x^2 + L_y^2 + L_z^2$ acts as $\ell(\ell + 1) \mathbb{I}$ (Lecture 10's proposition, whose proof used only brackets), so on every $\mathcal{Y}_\ell$ — and hence, by the completeness theorem below, on a dense subspace of $L^2(S^2)$ —

$$-\Delta_{S^2} = L_x^2 + L_y^2 + L_z^2.$$

The spherical Laplacian is the Casimir of the rotation group; the Laplace operator has been doing representation theory since the eighteenth century.

Theorem 11.14 (Spherical harmonics are complete). The spaces $\mathcal{Y}_\ell$ are mutually orthogonal in $L^2(S^2)$, and their closed span is everything:

$$L^2(S^2) = \widehat{\bigoplus_{\ell \geq 0} \mathcal{Y}_\ell}, \quad (11.4)$$

so the functions $Y_{\ell m}$, $\ell \geq 0$, $-\ell \leq m \leq \ell$, are an orthonormal basis of $L^2(S^2)$.

Proof. The rotation action on $L^2(S^2)$ is unitary: the surface measure is rotation-invariant (substitution). Let $\mathcal{P}_N$ be the space of restrictions to S^2 of polynomials of degree $\leq N$: finite-dimensional, invariant, and — by Proposition 11.8 with $r \equiv 1$ on the sphere — equal to $\sum_{\ell \leq N} \mathcal{Y}_\ell$. On $\mathcal{P}_N$ we have an honest finite-dimensional unitary representation, so its generators are anti-Hermitian (Lecture 9), the Casimir L^2 is Hermitian, and the $\mathcal{Y}_\ell$ are its eigenspaces with the distinct eigenvalues $\ell(\ell + 1)$: mutually orthogonal, with direct sum $\mathcal{P}_N$. Inside each $\mathcal{Y}_\ell$ the ladder proposition of Lecture 10 makes the $Y_{\ell m}$ orthonormal. Density: the restricted polynomials form a subalgebra of $C(S^2)$ containing the constants, separating points (the coordinates do) and closed under conjugation, hence uniformly dense by Stone–Weierstrass — imported here from the analysis course — and uniform density implies L^2 -density on a finite measure space, with $C(S^2)$ dense in $L^2(S^2)$ from functional analysis. An orthonormal family with dense span is a basis. ■

The physics writes itself. A particle in a central potential, $H = -\frac{1}{2\mu}\Delta + V(r)$, commutes with rotations (Lemma 11.6); separating $\psi = f(r)Y_{\ell m}$ and using Theorem 11.12 (which gives $\Delta_{S^2}Y_{\ell m} = -\ell(\ell + 1)Y_{\ell m}$) and the polar form $\Delta(f(r)Y) = (f'' + \frac{2}{r}f')Y + r^{-2}f\Delta_{S^2}Y$,

$$-\frac{1}{2\mu}\left(f'' + \frac{2}{r}f'\right) + \left(V(r) + \frac{\ell(\ell+1)}{2\mu r^2}\right)f = E f :$$

the centrifugal barrier of every quantum mechanics course is a Casimir eigenvalue. Each radial solution arrives $(2\ell + 1)$ -fold degenerate — Lecture 9's degeneracy doctrine with the dimensions now computed — and *only* integer ℓ occurs, by Theorem 11.5. Hydrogen's spectrum is degenerate beyond this, n^2 rather than $2\ell + 1$: a starred mystery (a hidden symmetry generated by the Runge–Lenz vector enlarges SO(3) to SO(4); optional project).

11.5 Euler angles and Wigner matrices

Computations with rotations need coordinates, and three angles suffice.

Proposition 11.15 (Euler angles). *Every $R \in \text{SO}(3)$ can be written*

$$R = R_z(\alpha) R_y(\beta) R_z(\gamma), \quad \alpha, \gamma \in [0, 2\pi), \beta \in [0, \pi]. \quad (11.5)$$

Proof. The unit vector $R\hat{z}$ has polar angles (β, α) for some $\beta \in [0, \pi]$, $\alpha \in [0, 2\pi)$ (Figure 11.3), and $R_z(\alpha)R_y(\beta)\hat{z}$ is exactly that vector. Hence $Q = (R_z(\alpha)R_y(\beta))^{-1}R$ fixes $\hat{z}$, so it preserves the xy -plane, restricts there to an element of SO(2), and equals $R_z(\gamma)$ for some γ . (At $\beta = 0$ or π only $\alpha \pm \gamma$ is determined — the chart has a seam, known to every gimbal; Sheet 11 examines it.) ■

Definition 11.16 (Wigner matrices). $D_{m'm}^j(U) = \langle j, m' | \Pi_j(U) | j, m \rangle$, the matrix elements of the irreducible representations in the ladder basis. Each D^j is a unitary $(2j + 1) \times (2j + 1)$ matrix with $D^j(UV) = D^j(U)D^j(V)$; for integer j , Theorem 11.5 makes D^ℓ a function on SO(3) as well. In Euler coordinates, with $U(\alpha, \beta, \gamma) = e^{-i\alpha J_z} e^{-i\beta J_y} e^{-i\gamma J_z}$,

$$D_{m'm}^j(\alpha, \beta, \gamma) = e^{-i(am' + \gamma m)} d_{m'm}^j(\beta), \quad d^j(\beta) = \text{matrix of } e^{-i\beta J_y}, \quad (11.6)$$

so the whole j -dependence sits in the single real matrix d^j (real because $J_\pm$ have real matrix elements in the ladder basis — Lecture 10's positive square roots — so $J_y = (J_+ - J_-)/2i$ is purely imaginary and $-i\beta J_y$ is real). For instance $d^{1/2}(\beta) = \begin{pmatrix} \cos \frac{\beta}{2} & -\sin \frac{\beta}{2} \\ \sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix}$, half-angles included.

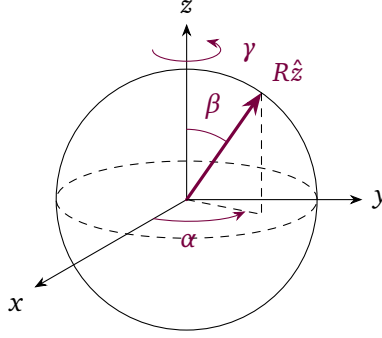


Figure 11.3: Euler angles. The pair (β, α) are the polar coordinates of $R\hat{z}$, the image of the north pole — β measured down from the z -axis, α around the equatorial plane from the x -axis (the dashed lines project $R\hat{z}$ onto that plane). These two angles fix $R_z(\alpha)R_y(\beta)$, which carries $\hat{z}$ to the right place; the leftover rotation about the original z -axis (upper circular arrow) is γ , the factor applied first in (11.5). At $\beta = 0$ or π the azimuth degenerates — the gimbal's seam.

One identity ties the matrices to Section 11.4. In the basis $\{Y_{\ell m}\}$ of $\mathcal{Y}_\ell$ the function action has matrix elements D^ℓ : $\Pi(R)Y_{\ell m} = \sum_{m'} D_{m'm}^\ell(R)Y_{\ell m'}$. Only the $m' = 0$ harmonic survives at the north pole — from $\Pi(R_z(\theta))Y_{\ell m'} = e^{-i\theta m'}Y_{\ell m'}$ evaluated at the fixed point $\hat{z}$ — so evaluating the action there gives

$$D_{0m}^\ell(R) = c_\ell^{-1} Y_{\ell m}(R^{-1}\hat{z}), \quad c_\ell = Y_{\ell 0}(\hat{z}) = \sqrt{\frac{2\ell+1}{4\pi}} \text{ (standard normalisation)} : \quad (11.7)$$

one row of every integer Wigner matrix is a spherical harmonic in disguise.

Example 11.17 (The spin-one Wigner matrix). For $j = 1$ the exponential series collapses exactly as it did for the Pauli matrices in Lecture 8: the spin-one J_y of Lecture 10 has eigenvalues $0, \pm 1$, so $J_y^3 = J_y$ and the series folds into $e^{-i\beta J_y} = \mathbb{1} - i \sin \beta J_y + (\cos \beta - 1) J_y^2$, giving

$$d^1(\beta) = \begin{pmatrix} \frac{1+\cos \beta}{2} & -\frac{\sin \beta}{\sqrt{2}} & \frac{1-\cos \beta}{2} \\ \frac{\sin \beta}{\sqrt{2}} & \cos \beta & -\frac{\sin \beta}{\sqrt{2}} \\ \frac{1-\cos \beta}{2} & \frac{\sin \beta}{\sqrt{2}} & \frac{1+\cos \beta}{2} \end{pmatrix},$$

rows and columns ordered $m = 1, 0, -1$. Three readings. First, this is no stranger: it is the ordinary rotation $R_y(\beta)$ rewritten in the harmonic basis $\{-(x+iy)/\sqrt{2}, z, (x-iy)/\sqrt{2}\}$ of H_1 (Lecture 10's ladder from $x+iy$ gives this basis up to normalisation and an overall sign; the signs shown are the standard spherical-basis convention) — spin one is the vector representation (Sheet 10) in spherical dress. One column suffices to check: $\Pi(R_y(\beta))z = x \sin \beta + z \cos \beta$, whose coordinates in that basis are exactly the middle column. Second, the middle row is the identity (11.7) at $\ell = 1$: for instance $D_{00}^1 = \cos \beta = c_1^{-1} Y_{10}(R^{-1}\hat{z})$, the polar angle of $R^{-1}\hat{z}$ being β ; the remaining entries follow in the same way from $Y_{1,\pm 1}$, and the Euler phases from (11.6). Third, the endpoints: $d^1(\pi)$ is the antidiagonal flip $m \mapsto -m$ with signs $(-1)^{1-m}$, and

$$d^1(2\pi) = \mathbb{1}, \quad \text{while} \quad d^{1/2}(2\pi) = -\mathbb{1} :$$

Lecture 10's $(-1)^{2j}$, now wearing Euler coordinates.

Everything else about these matrices is computational and lives in Notebook 11 (WignerD) and on Sheet 11; Lecture 12 will need only their traces.

11.6 Rays, projective representations, and Wigner's theorem

One debt remains, and it is conceptual. Lecture 10 took the symmetry group of a quantum system to be $SU(2)$ while geometry insisted on $SO(3)$, and promised this lecture would arbitrate. The arbitration turns on what a quantum state actually is.

A state is not a vector. Two unit vectors differing by a phase, ψ and $e^{i\alpha}\psi$, are indistinguishable by any measurement: every prediction of quantum mechanics is built from transition probabilities

$$P([\varphi], [\psi]) = |\langle \varphi | \psi \rangle|^2,$$

which depend only on the rays $[\psi] = \{e^{i\alpha}\psi : \alpha \in \mathbb{R}\}$. The honest state space is the set of unit rays, and the honest notion of symmetry follows: a *quantum symmetry* is a bijection T of the rays preserving all transition probabilities. What such maps can be is settled by a theorem we import — a conceptual import on which the rest of the course rests.

Theorem 11.18 (Wigner's theorem; imported). *Every quantum symmetry T is implemented by an operator $U: \mathcal{H} \rightarrow \mathcal{H}$ with $T[\psi] = [U\psi]$, where U is either unitary or antiunitary — additive, conjugate-linear, with $\langle U\varphi | U\psi \rangle = \overline{\langle \varphi | \psi \rangle}$ — and, when $\dim \mathcal{H} \geq 2$, U is unique up to a phase.*

Reference for the omitted proof: Weinberg, *The Quantum Theory of Fields I*, Appendix 2.A.

Antiunitary implementers are not exotica — time reversal is one: it must conjugate the phases e^{-iEt} to reverse motion, so it cannot be linear — but they cannot occur in a connected symmetry group: every element of a connected matrix Lie group is a finite product of exponentials (Lecture 8), hence of squares $e^X = (e^{X/2})^2$, and the implementer of a square, $U_{h^2} = \omega U_h U_h$, is unitary whether U_h is unitary or antiunitary. For rotations, then, only unitaries — but unitaries defined up to phases, and the phases are the whole story.

Definition 11.19 (Projective representation). *A projective unitary representation of a group G on $\mathcal{H}$ is a map $g \mapsto U_g$ into the unitaries with*

$$U_g U_h = \omega(g, h) U_{gh}, \quad |\omega(g, h)| = 1 : \quad (11.8)$$

a representation up to phases — exactly what Wigner's theorem delivers when a group acts by quantum symmetries.

Proposition 11.20 (Half-integer spin is projectively lawful). *For every j , choose for each $R \in SO(3)$ a preimage $U_R \in \Phi^{-1}(R)$ and set $W_R = \Pi_j(U_R)$. Then*

$$W_R W_{R'} = \pm W_{RR'} :$$

a projective unitary representation of $SO(3)$, whose action on rays is independent of all choices. For integer j the signs can be removed (Theorem 11.5); for half-integer j they cannot: no honest representation of $SO(3)$ induces the same ray action.

Proof. Since $U_R U_{R'} \in \Phi^{-1}(RR') = \{\pm U_{RR'}\}$ and $\Pi_j(-U) = (-1)^{2j} \Pi_j(U)$ (Lecture 10), the multiplication rule holds with signs, and replacing U_R by $-U_R$ changes W_R by at most a sign — invisible on rays. Suppose now an honest representation $\tilde{\Pi}$ of $SO(3)$ induced the same ray action. For each $U \in SU(2)$ the unitaries $\tilde{\Pi}(\Phi(U))$ and $\Pi_j(U)$ implement the same ray map, so by the uniqueness in Wigner's theorem $\tilde{\Pi}(\Phi(U)) = \lambda(U) \Pi_j(U)$ for a phase $\lambda(U)$; being a ratio of homomorphisms, λ is a homomorphism $SU(2) \rightarrow U(1)$, continuous because $\lambda(U) = \frac{1}{2^{j+1}} \text{tr}(\tilde{\Pi}(\Phi(U)) \Pi_j(U)^{-1})$ is. Its derivative (Lecture 9's differentiation theorem,

applied to this one-dimensional representation) is a Lie algebra map $\mathfrak{su}(2) \rightarrow \mathfrak{u}(1)$ into an abelian algebra, and it must kill all brackets — but brackets span $\mathfrak{su}(2)$ ($X_1 = [X_2, X_3]$ and cyclically), so the derivative vanishes and $\lambda \equiv 1$ on the connected $SU(2)$. Then $\tilde{\Pi} \circ \Phi = \Pi_j$, forcing $\Pi_j(-\mathbb{K}) = \tilde{\Pi}(\mathbb{K}) = \mathbb{K}$ — false for half-integer j . ■

Here, at last, is the arbitration. Quantum mechanics is formulated on rays; its symmetry groups act by projective representations; and the projective representations of $SO(3)$ are precisely the honest representations of $SU(2)$ in which $-\mathbb{K}$ acts as a scalar (the irreducible ones, and sums of irreducibles of a fixed parity of $2j$; for $\Pi_0 \oplus \Pi_{1/2}$ the two lifts $\pm U$ of a rotation give different ray maps) — the half we proved above, the converse (every projective representation of $SO(3)$ arises so, after adjusting phases) being Bargmann's theorem, stated as Theorem 11.21 below and imported. *The physicist's rotation group is $SU(2)$* , not by decree but because the quantum kinematics of rays makes the simply connected cover available free of charge. Lecture 10's IOU is paid in full: a spin- $\frac{1}{2}$ system violates nothing; it merely uses the phases that vectors carry and rays forgive. (Not every group is so lucky: for the Galilei group genuinely irremovable phases appear, carrying the particle's *mass* — close kin of the central element of Lecture 13's Heisenberg algebra. For the Lorentz group, Lecture 14 will find the same luck as here.)

Theorem 11.21 (Bargmann; imported). *Let G be a connected matrix Lie group whose Lie algebra $\mathfrak{g}$ satisfies a condition we do not define — $H^2(\mathfrak{g}, \mathbb{R}) = 0$, the vanishing of its second cohomology — which holds in particular for $\mathfrak{su}(2)$ and for every semisimple $\mathfrak{g}$ (Whitehead's lemma; neither term is developed in this course). Then every projective unitary representation of G — every assignment $g \mapsto W_g$, continuous in the sense that the ray map $[\psi] \mapsto [W_g \psi]$ depends continuously on g , obeying $W_g W_{g'} = c(g, g') W_{gg'}$ with phases $c(g, g') \in U(1)$ — is, on rays, induced by an honest strongly continuous unitary representation of the universal cover $\tilde{G}$: the phases $c(g, g')$ can be removed after lifting to $\tilde{G}$. Here $\tilde{G}$ is the universal cover described in Lecture 9's monodromy remark; for $G = SO(3)$ it is $SU(2)$.*

Reference for the omitted proof: Bargmann, *On unitary ray representations of continuous groups*, Ann. of Math. **59** (1954) 1–46; Weinberg, *The Quantum Theory of Fields I*, §2.7.

Applied to $SO(3)$, whose universal cover is $SU(2)$ with $\mathfrak{su}(2)$ semisimple, this supplies exactly the converse used above: every projective representation of $SO(3)$ arises, after adjusting phases, from an honest representation of $SU(2)$. (The Galilei group fails the hypothesis — $H^2 \neq 0$, the surviving cocycle being the mass — which is why its phases do not lift. For the connected Lorentz group the same argument uses the semisimple Lorentz algebra and lifts to its $SL(2, \mathbb{C})$ cover, as Lecture 14 explains. The full Poincaré algebra is not semisimple; its distinct cohomological lifting input is stated at the point of use in Lecture 15.)

And the minus sign is not philosophy. In 1975 two independent crystal-neutron-interferometer experiments made it relative and therefore observable. Rauch and collaborators coherently split unpolarised slow neutrons, varied the unequal magnetic-field integrals along the two paths, and observed output-intensity oscillations with the predicted 4π spinor period. Werner and collaborators instead used a polarised beam and measured the interferometric phase shift produced by magnetic precession, directly identifying the sign reversal after a 2π rotation. Both experiments use magnetic precession and path recombination, but they are not the same protocol. The $(-1)^{2j}$ of Lecture 10, computed from bare algebra, is visible in a reactor hall.

Primary experimental sources: H. Rauch et al., “Verification of coherent spinor rotation

of fermions,” *Physics Letters A* **54** (1975), 425–427, doi:10.1016/0375-9601(75)90798-7; S. A. Werner et al., “Observation of the phase shift of a neutron due to precession in a magnetic field,” *Physical Review Letters* **35** (1975), 1053–1055, doi:10.1103/PhysRevLett.35.1053.

The season’s geography is complete: one algebra, two groups, a two-to-one bridge between them, and quantum mechanics choosing the upper deck. Lecture 12 restarts the engines of Part I — invariant integration in place of group sums, characters (including the crystal-field formula Part I borrowed on credit), and the question the experimentalists ask daily: what happens when two angular momenta meet. Lecture 14 will replay today’s construction one signature up, with $SL(2, \mathbb{C})$ standing over the Lorentz group.

Remark 11.22 (Computer algebra). Notebook 11 checks `SphericalHarmonicY` and `WignerD` against our ladder constructions; verifies $\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1)Y_{\ell m}$ numerically; plots $|Y_{\ell m}|$ over the sphere; and animates the double cover — a path of $SU(2)$ matrices beside its image rotations, the 2π loop arriving at $-\mathbb{I}$ while the 4π loop closes — with the belt trick rendered on screen.

Summary.

- *Double cover.* $\Phi : SU(2) \rightarrow SO(3)$ is a two-to-one homomorphism with kernel $\{\pm \mathbb{I}\}$ (Theorem 11.1); in $SO(3)$ the loop ρ is not contractible but ρ^2 is (Proposition 11.3) — in the language of topology, not developed here, $SO(3)$ has fundamental group $\mathbb{Z}/2$.
- *Descent.* The representations of $SO(3)$ are exactly the integer-spin representations of $SU(2)$ (Theorem 11.5).
- *Spherical harmonics.* The degree- ℓ harmonic polynomials carry the spin- ℓ representation, of dimension $2\ell + 1$ (Theorem 11.9, Proposition 11.8), with spherical-Laplacian eigenvalue $-\ell(\ell + 1)$ (Theorem 11.12).
- *Completeness.* The $Y_{\ell m}$ form an orthonormal basis of $L^2(S^2)$ (Theorem 11.14); Euler angles and the Wigner matrices coordinatise rotations (Proposition 11.15).
- *Projective representations.* Quantum states are rays (Wigner’s theorem, Theorem 11.18), so half-integer spin is lawful and $SU(2)$ is the operative group (Proposition 11.20).

Tutorial. Exercise Sheet 11.

References

Hall §4.7 ($SO(3)$ versus $SU(2)$); Stillwell Ch. 2; Tung (the $SO(3)$ and $SU(2)$ – $SO(3)$ chapters); Elliott–Dawber (rotation-group sections); Woit (rotations and spin); Costa–Fogli Ch. 2. For Bargmann’s theorem (Theorem 11.21): Bargmann, *Ann. of Math.* **59** (1954) 1–46; Weinberg, *QFT I* §2.7. For the 1975 neutron experiments: Rauch et al., *Phys. Lett. A* **54**, 425–427; Werner et al., *Phys. Rev. Lett.* **35**, 1053–1055.

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